

Challenge 4: Sidorenko Conjecture for Half-Graphs

Definition 1 (Graph homomorphism density). Let G and H be simple graphs.

1. A function $f : V(H) \rightarrow V(G)$ is a *graph homomorphism* from H to G if for each edge $uv \in E(H)$, we have $f(u)f(v) \in E(G)$.
2. Let $\text{hom}(H, G)$ be the number of graph homomorphisms $H \rightarrow G$.
3. The *homomorphism density* of H in G is the fraction

$$t_H(G) := \frac{\text{hom}(H, G)}{|V(G)|^{|V(H)|}}$$

Conjecture 1.1 (Sidorenko's Conjecture [1]). For every bipartite graph H and for any graph G , we have:

$$t_H(G) \geq t_{K_2}(G)^{|E(H)|}$$

We are interested in the special case of the conjecture, where $H = H_r$ is a half-graph with $2r$ vertices, as follows. As far as we are aware, the conjecture has not been asked anywhere for half-graphs, and the only reason we do it for half-graphs here is that we believe that they are somewhat generic, and it will help us formulate an easier scalable version. (TODO really ask some experts in extremal combinatorics here)

Definition 2 (Half-graph). Let $A_r = \{a_1, \dots, a_r\}$ and $B_r = \{b_1, \dots, b_r\}$ be disjoint sets of size r . A *half-graph* is the bipartite graph on the vertex set $A_r \cup B_r$ with the following edges present: $E = \{a_i b_j : i \leq j\}$ (Note that these are about 'half' of the possible edges between A_r and B_r , hence the name).

Conjecture 1.2 (Scaled weakening of Sidorenko's Conjecture). Let $r \in \mathbb{N}$. For any graph G , we have:

$$t_{H_r}(G) \geq t_{K_2}(G)^{(r^2+r)/2}$$

TODO find out for which half-graphs it is known.

References

- [1] A. F. Sidorenko. “A Correlation Inequality for Bipartite Graphs”. In: *Graphs and Combinatorics* 9.2–4 (1993), pp. 201–204. DOI: 10.1007/BF02988307. URL: <https://doi.org/10.1007/BF02988307>.